

Contact Info Gathering Flow

Overview

Before sending notifications, the system must collect and store user contact information.

The notification system requires:

- Mobile device tokens
- Phone numbers
- Email addresses

These details are necessary to deliver notifications through different channels.

Contact Information Collection Flow

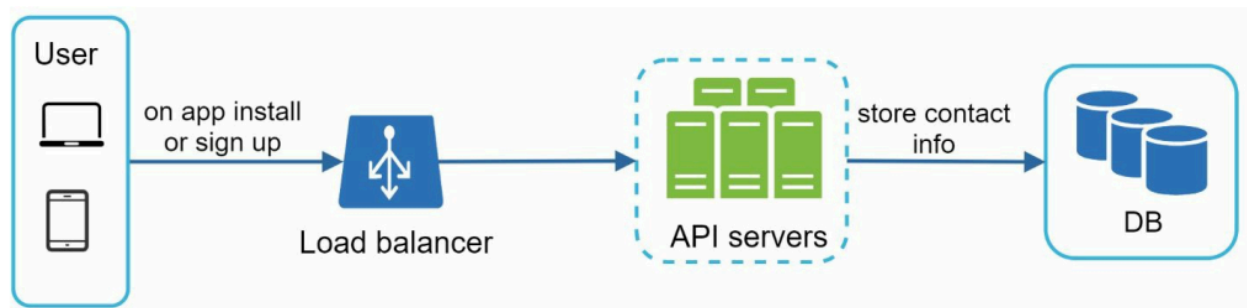


Figure illustrates how user contact information is collected and stored.

Flow Explanation

Step 1: User Installs App / Signs Up

- User installs the application
- Or creates an account for the first time

Step 2: API Server Collects User Information

The API server gathers:

- Email address
- Phone number
- Device token

Step 3: Store Information in Database

Collected contact details are stored in database tables for future notification delivery.

Database Design

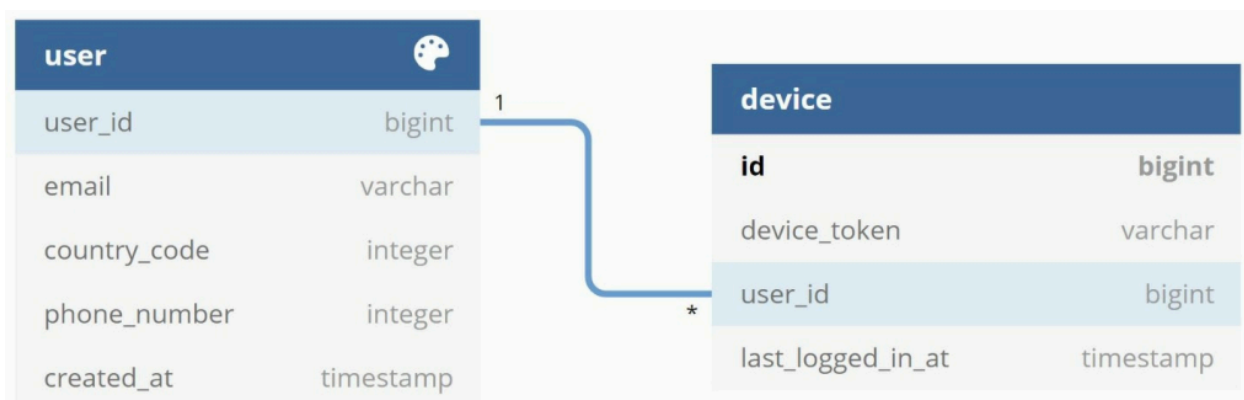


Figure shows the simplified database schema for storing contact information.

User Table

Stores

- User profile information
- Communication details

Example Fields

Field	Description
user_id	Unique user identifier
email	User email address
phone_number	User phone number

Device Table

Stores

- Mobile device information
- Push notification tokens

Example Fields

Field	Description
device_id	Unique device identifier
user_id	Associated user
device_token	Push notification token
device_type	iOS / Android

Important Design Insight

One User → Multiple Devices

A user may own multiple devices:

- Smartphone
- Tablet
- Laptop

Implication

- Push notifications can be sent to:
 - All registered devices of a user

Example

None

User A

└─ iPhone

└─ Android Tablet

└─ Laptop

Why Device Tokens are Important?

Purpose

- Device token uniquely identifies a device
- Used by:
 - APNS (iOS)
 - FCM (Android)

Without Device Token

- Push notification cannot be delivered

Benefits of This Design

1. Multi-device Support

- Notifications reach all user devices

2. Centralized Contact Management

- Easy retrieval of contact information

3. Scalability

- Supports millions of users and devices

4. Flexibility

- Supports multiple notification channels:
 - Push
 - SMS
 - Email

Key Takeaway

The contact info gathering system is responsible for:

- Collecting user communication details
- Storing device tokens
- Supporting multi-device notifications

Main Storage Design:

Table	Stores
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User Table	Email + Phone Number
Device Table	Device Tokens

Goal:

Enable reliable delivery of:

- Push notifications
- SMS messages
- Emails

to all registered user devices.